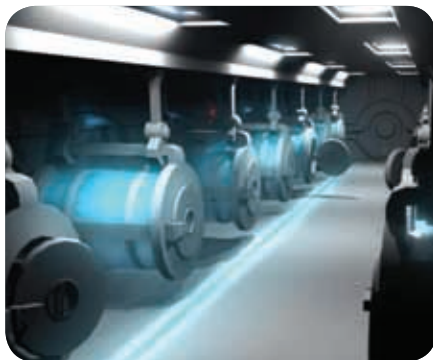


cial light, air and water filtration systems; it's mind boggling. Add to it, the effects of Zero G on the human body. Over prolonged periods of microgravity, astronauts have suffered loss of bone density, decreased muscle strength and even difficulty in breathing. Over time astronauts simply put, become clumsy. Imagine what effects that'll have on the success of a mission. Also, while the ship's hull can protect a crew from the sub-zero temperatures outside, it can't protect them much from cosmic radiation. There is a whole spectrum of radiation out there which our bodies will not react kindly to. In short the endless vacuum will keep reminding those on board that they have no business being there.

SPEED

The fastest that anything can travel in the known universe is at the speed of light (3×10^8 m/s). If we eventually build a spacecraft that can indeed travel at that speed, it will still take hundreds (and even thousands) of years to reach places of significance. That's several lifetimes in human terms. How do we overcome this? Proposed solutions to both the distance, time and life-support problems include hypersleep or cryogenic freezing of the human body i.e. putting the body in a state of suspended animation. All cell functions are reduced



to a minimum level and the body is then revived when the destination is reached. The other solution is actually having generations of humans breed and procreate on board a ship. That is of course if postural instability from Zero G and rays of cosmic radiation don't render them infertile first! Maintaining a functioning society will also be problematic. Imagine the revolts, and the hippies trying to disrupt a mission!

THE VALUE WE ATTACH TO HUMAN LIFE

A side effect of the human condition, is that we are creatures that value life! A hive mind on the other hand wouldn't think twice about sacrificing the life of a few drones for the greater good of the whole. If humans are ever to start taking steps towards the great abyss above, be sure we will face many failed missions and loss of life. The two biggest space exploration disasters (Challenger in 1986 and Columbia in 2003) collectively claimed the lives of fourteen astronauts. A small price to pay? Not many think so...

PANDORUM

As a result of being confined to a fixed place sailors on long voyages at sea often showed symptoms of a peculiar kind of lunacy called Cabin Fever. The symptoms of this disorder included irrational behaviour, restlessness, frustration, uncontrollable laughter and even a desire to jump overboard. Of course if they did jump overboard these wayward sailors were rescued and put in the brig. In space you don't have that option. A 2009 movie called Pandorum explores this concept which is



a very likely possibility on space voyages that last many generations.

THE EXPENSE

Even if all the world's countries forget their silly squabbles (over who owns a strip of land or which politician should resign) we would still not have the resources to build something that can carry people to the nearest solar system in over a hundred years. To give you an idea of how expensive space flight can be consider this fact: a shuttle launch today costs \$1.3 billion. It has been argued that if every square inch of cargo carried by space shuttles converted miraculously to gold on its return journey, the mission would still lose money.

HOW WILL WE DO IT THEN?

Considering all the problems we've listed so far you wouldn't be wrong to think the era of manned space exploration came



to an end with mankind landing on the moon. The next leaps, which are exponentially larger, are not within our foreseeable reach. And so we have no option than to turn to our own creations – machines and robots – in our effort to reach for the stars. The comparatively lower cost and success of the two Mars rover missions are a clear indicator of the way forward. Even if we consider hypersleep for populating far away planets, robots will be our caretakers while we dream of life amongst the stars. Mankind has no option but to engage in deep space exploration by proxy, for a long time to come. **d**



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